1 Caesar Cipher Algorithm

This is a simple substitution cipher where each letter in plaintext is shifter a number of position down (or up) the alphabet. The key is numerical and wraps around A - Z (or vice-versa). To encrypt you shift letter forward while, decryption reverses this by shifting letters backward.

Example:
Suppose we have a plaintext (P_i) as Hello and a key ($k = 3$), then the encryption given by the following formula:

$$C_i = (P_i + k) \quad MOD \quad 26$$

yields:
"H" is at index position 7 counting from 0, "e" at position 4 and the last digit "o" at position 14. Shifting them three position forward we will have a resultant index of $10, 7, ..., 17$ respectively.

The final step is to compute the modulus and return the resultant letters. e.g: 10 MOD 26 = 10 so, the resultant index being 10, then, position alphabet will be "K". This process is repeated for all the characters to completion.

2 Vigenere Cipher Algorithm

Vigenere Cipher is a poly-alphabetic substitution cipher. This algorithm is stronger compared to Caesar Cipher one. Each letter in the message(also plaintext) use different shifting key. The shift is based on a secret key which is a text.
The encryption is computed as follows:

$$C_i = (P_i + key) \quad Mod \quad 26$$

. Remember:

1. C_i = Ciphered text
2. P_i = Plaintext letter at position (i)
3. **Mod 26** = Ensures there is wrapping

Note: The direction matters because, for encryption we use +1 and decryption we can use -1.